

VI

DeFi DEEP DIVE

DeFi can be loosely broken into sectors based on the functionality type of the dApp. Many dApps could fit into multiple categories, so we attempt to place them into the most relevant category. We examine DeFi platforms in the taxonomy of lending/credit facilities, DEXes, derivatives, and tokenization.¹ We mainly focus on the Ethereum network due to its popularity, but DeFi innovations are occurring on many blockchains including Stellar and EOS.² Polkadot³ is another platform that employs a type of proof-of-stake consensus.

CREDIT/LENDING

MakerDAO

MakerDAO⁴ (DAO is a *decentralized autonomous organization*) is often considered an exemplar of DeFi. For a series of

applications to build on each other, there must necessarily be a foundation. The primary value-add of MakerDAO is the creation of a cryptocollateralized stablecoin, pegged to USD. This means the system can run completely from within the Ethereum blockchain without relying on outside centralized institutions to back, vault, and audit the stablecoin. MakerDAO is a two-token model where a governance token MKR yields voting rights on the platform and participates in value capture. The second token is a stablecoin called DAI – a staple token in the DeFi ecosystem with which many protocols integrate, including a few we will discuss later.

DAI is generated as follows. A user can deposit ETH or other supported ERC-20 assets into a *vault*, which is a smart contract that escrows collateral and keeps track of the USD-denominated value of the collateral. The user can then mint DAI up to a certain collateralization ratio on their assets. This creates a “debt” in DAI that the vault holder must pay back. The DAI is the corresponding asset that can be used any way the vault holder wishes. For example, the user can sell the DAI for cash or leverage it into more of the collateral asset⁵ and repeat the process. Due to the volatility of ETH and most collateral types, the collateralization requirement is far in excess of 100 percent and usually in the 150–200 percent range.

The basic idea underlying the DAI mechanism is not new; it is simply a collateralized debt position. For example, a homeowner in need of some liquidity can pledge their house as collateral to a bank and receive a mortgage loan structured to include a cash takeout. The price volatility of ETH is much

greater than for a house; as such, collateralization ratios for the ETH–DAI contract are higher than a traditional mortgage. In addition, no centralized institution is necessary because everything happens within the Ethereum blockchain.

Let's consider a simple example. Suppose an ETH owner needs liquidity but does not want to sell her ETH because she thinks it will appreciate. The situation is analogous to the homeowner who needs liquidity but does not want to sell their house. Let's say an investor has 5 ETH at a market price of \$200 (total value of \$1,000). If the collateralization requirement is 150 percent, then the investor can mint up to 667 DAI ($\$1,000/1.5$ with rounding). The collateralization ratio is set high to reduce the probability that the loan debt exceeds the collateral value. In addition, for the DAI token to be credibly pegged to the USD, the system needs to avoid the risk that the collateral is worth less than $\$1 = 1$ DAI.

Given the collateralization ratio of 1.5, it would be unwise to mint the 667 DAI because if the ETH ever dropped below \$200 the contract would be undercollateralized – the equivalent of a *margin call*. We are using traditional finance parlance, but in DeFi there is no communication from your broker about the need to post additional margin or to liquidate the position and also no grace period. Liquidation can happen immediately.

As such, most investors choose to mint less than 667 DAI to give themselves a buffer. Suppose the investor mints 500 DAI, which implies a collateralization ratio of 2.0 ($\$1,000/2.0 = 500$). Let's explore two scenarios. First, suppose the price of ETH rises by 50 percent so that the

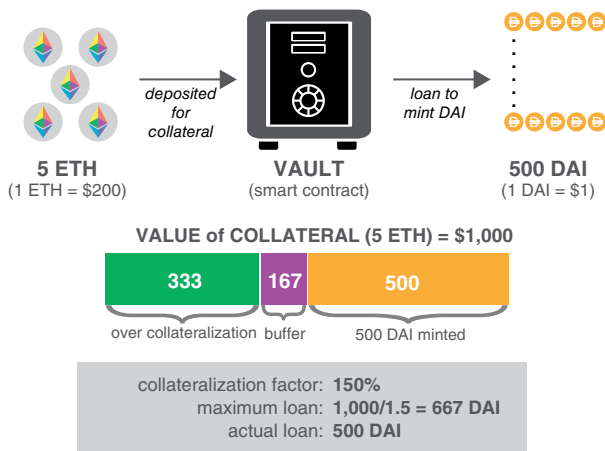
collateral is worth \$1,500. Now, the investor can increase the size of their loan. To maintain the collateralization of 200 percent, the investor can mint an extra 250 DAI.

A more interesting scenario is when the value of the collateral drops. Suppose the value of the ETH drops by 25 percent, from \$200 to \$150. In this case, the value of the collateral drops to \$750 and the collateralization ratio drops to 1.5 ($\$750/1.5 = 500$).

The vault holder faces three scenarios. First, they can increase the amount of collateral in the contract (by, e.g., adding 1 ETH). Second, they can use the 500 DAI to pay back the loan and repatriate the 5 ETH. These ETH are now worth \$250 less, but the depreciation in value would have happened regardless of the loan. Third, the loan is liquidated by a *keeper* (any external actor) who is incentivized to find contracts eligible for liquidation. The keeper auctions the ETH for enough DAI to pay off the loan. In this case, 3.33 ETH would be sold and 1.47 would be returned to the vault holder (the keeper earns an incentive fee of 0.2 ETH). The vault holder then has 500 DAI worth \$500 and 1.47 ETH worth \$220. This analysis does not include gas fees.

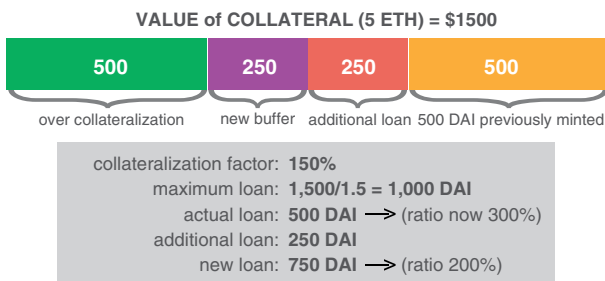
Two forces in this process reinforce the stability of DAI: overcollateralization and market actions. In the liquidation, ETH are sold and DAI are purchased, which exerts positive price pressure on DAI. This simple example does not address many features in the MakerDAO ecosystem (Figure 6.1), in particular the fee mechanisms and the debt limit, which we will now explore.

DeFi Deep Dive



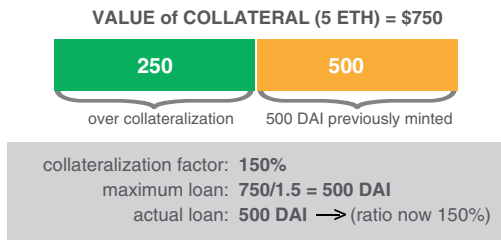
Scenario 1

ETH appreciates 50% \$200 → \$300



Scenario 2

ETH depreciates 25% \$200 → \$150



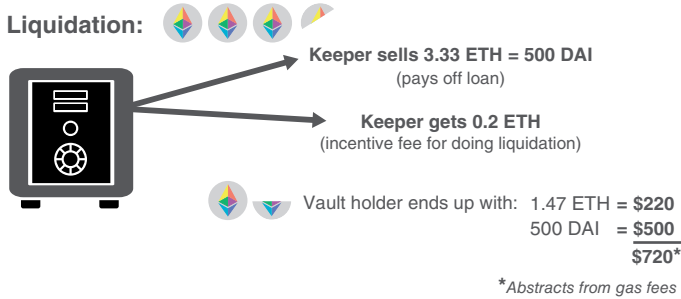


Figure 6.1 The Mechanics of MakerDAO's DAI

The viability of the MakerDAO ecosystem critically depends on DAI maintaining a 1:1 peg to the USD. Various mechanisms are in place to incentivize demand and supply to drive the price toward the peg. The primary mechanisms for maintaining the peg are the debt ceiling, stability fee, and DAI savings rate (DSR). These parameters are controlled by holders of the governance token Maker (MKR) and MakerDAO governance, which we will discuss toward the end of this section.

The stability fee is a variable interest rate that vault holders pay in DAI on any DAI debt they generate. The interest rate can be raised or lowered (even to a negative value) to incentivize the generation or repayment of DAI to drive its price toward the peg. The stability fee funds the DSR, a variable rate any DAI holder can earn on their DAI deposit. The DSR compounds on a per-block basis. The stability fee, which must always be greater or equal to the DSR, is enforced by the smart contracts powering the platform. The analogous situation in centralized finance is that the loan

rate is always higher than the deposit rate. Lastly, a smart contract–enforced DAI debt ceiling can be adjusted to allow for more or less supply to meet the current level of demand. If the protocol is at the debt ceiling, no new DAI is able to be minted in new vaults until the old debt is paid or the ceiling is raised.

To stay above the liquidation threshold, a user can deposit more collateral into the vault to keep the DAI safely collateralized. When a position is deemed to be under the liquidation ratio, a keeper can initiate an auction (i.e., sell some of the ETH collateral⁶) to liquidate the position and close the vault holder's debt. The *liquidation penalty* is calculated as a percentage of the debt and is deducted from the collateral in addition to the amount needed to close the position.

After the auction, any remaining collateral reverts to the vault owner. The liquidation penalty acts as an incentive for market participants to monitor the vaults and to trigger an auction when a position becomes undercollateralized. If the collateral drops so far in value that the DAI debt cannot be fully repaid, the position is closed and the protocol accrues *protocol debt*. A buffer pool of DAI exists to cover it up to a certain amount. The solution involves the governance token MKR and the governance system.

The MKR token controls MakerDAO. Holders of the token have the right to vote on protocol upgrades, including supporting new collateral types and tweaking parameters such as collateralization ratios. MKR holders are expected to make decisions in the best financial interest of the platform.

Their incentive is that a healthy platform should increase the value of their share in the platform's governance. For example, because of poor governance, the buffer pool could be insufficient to pay back the protocol debt. If all other measures to repay the debt have failed, *global settlement* is a safety mechanism that can be used in which newly minted MKR tokens are auctioned off in exchange for DAI and the DAI are used to repay the debt. Global settlement dilutes the MKR share, so stakeholders are incentivized to avoid it and keep protocol debt to a minimum.

MKR holders are collectively the owners of the future of MakerDAO. A proposal and corresponding approved vote can change any of the parameters available on the platform. Other possible parameter changes include supporting new collateral types for vaults and adding upgrades to functionality. MKR holders could, for instance, vote to pay themselves a dividend funded by the spread between the interest payments paid by vault holders and the DAI savings rate. The reward of receiving this dividend would need to be weighed against any negative community response (e.g., a backlash against rent seeking from a previously no-rent protocol) that might decrease the value of the protocol and the MKR token.

A number of features make DAI attractive to users. Importantly, users can purchase and use DAI without having to generate it in a vault – they can simply purchase DAI on an exchange without needing to know the underlying mechanics of how they are created. Holders can easily earn

the DAI savings rate by using the protocol, and more technologically and financially sophisticated users can use the MakerDAO web portal to generate vaults and create DAI to get liquidity from their assets without having to sell them. It is easy to sell DAI and purchase an additional amount of the collateral asset to get leverage.

A noteworthy drawback to DAI is that its supply is always constrained by demand for ETH-collateralized debt. No clear arbitrage loop exists to maintain the peg. For example, the stablecoin USDC is always redeemable with no fees by Coinbase for \$1. Arbitrageurs have a guaranteed (assuming solvency of Coinbase) strategy in which they can buy USDC at a discount or sell it at a premium elsewhere and redeem on Coinbase. This is not true for DAI. Irrespective of any drawbacks, the simplicity of DAI makes it an essential building block for other DeFi applications (Table 6.1).

Table 6.1 The Problems That MakerDAO Solves

Traditional	
Finance Problem	MakerDAO Solution
<i>Centralized control:</i> Interest rates are influenced by the U.S. Federal Reserve and access to loan products controlled by regulation and institutional policies.	MakerDAO platform is openly controlled by the MKR holders.
<i>(Continued)</i>	

Table 6.1 (Continued)

Traditional	
Finance Problem	MakerDAO Solution
<i>Limited access:</i> Obtaining loans is difficult for a large majority of the population.	Open ability to take out DAI liquidity against an overcollateralized position in any supported ERC-20 token. Access to a competitive USD-denominated return in the DSR.
<i>Inefficiency:</i> Acquiring a loan involves costs of time and money.	Instant liquidity at the push of a button with minimal transaction costs.
<i>Lack of interoperability:</i> Cannot trustlessly use USD or USD-collateralized token in smart contract agreements.	Issuance of DAI, a permissionless USD-tracking stablecoin backed by cryptocurrency. DAI can be used in any smart contract or DeFi application.
<i>Opacity:</i> Unclear collateralization of lending institutions.	Transparent collateralization ratios of vaults visible to entire ecosystem.

Compound

Compound is a lending market that offers several different ERC-20 assets for borrowing and lending. All the tokens in

a single market are pooled together so every lender earns the same variable rate, and every borrower pays the same variable rate. The concept of a credit rating is irrelevant, and because Ethereum accounts are pseudonymous, enforcing repayment in the event of a loan default is virtually impossible. For this reason, all loans are overcollateralized in a collateral asset different from the one being borrowed. If a borrower fall below their collateralization ratio, their position is liquidated to pay back their debt. The debt can be liquidated by a keeper, similar to the process used in MakerDAO vaults. The keeper receives a bonus incentive for each unit of debt they close out.

The collateralization ratio is calculated via a *collateral factor*. Each ERC-20 asset on the platform has its own collateral factor ranging from 0 to 90 percent. A collateral factor of zero means an asset cannot be used as collateral. The required collateralization ratio for a single collateral type is calculated as 100 divided by the collateral factor. Volatile assets generally have lower collateral factors, which mandate higher collateralization ratios due to increased risk of a price movement that could lead to undercollateralization. An account can use multiple collateral types at once, in which case the collateralization ratio is calculated as 100 divided by the weighted average of the collateral types by their relative sizes (denominated in a common currency) in the portfolio.

The collateralization ratio is similar to a reserve multiplier in traditional banking, constraining the amount of

“borrowed” dollars that can be in the system relative to the “real” supply. For instance, there is occasionally more DAI in Compound than is actually supplied by MakerDAO because users are borrowing and resupplying or selling to others who resupply. Importantly, all MakerDAO supply is ultimately backed by real collateral, and there is no way to borrow more collateral value than has been supplied.

For example, suppose an investor deposits 100 DAI with a collateral factor of 90. This transaction alone corresponds to a required collateralization ratio of 111 percent. Assuming 1 DAI = \$1, the investor can borrow up to \$90 worth of any other asset in Compound. If he or she borrows the maximum and the price of the borrowed asset increases at all, the position is subject to liquidation. Suppose the investor also deposit two ETH with a collateral factor of 60 and a price of \$200/ETH. The total supply balance is now \$500, with 80 percent being ETH and 20 percent being DAI. The required collateralization ratio is $100/(0.8*60 + 0.2*90) = 151$ percent (Figure 6.2).

The supply and borrow interest rates are compounded every block (approximately 15 seconds on Ethereum producing near continuous compounding) and are determined by the utilization percentage in the market. Utilization is calculated as total borrow/total supply. The utilization rate is used as an input parameter to a formula that determines the interest rates. The remaining parameters are set by *Compound Governance*, which we describe near the end of this section.

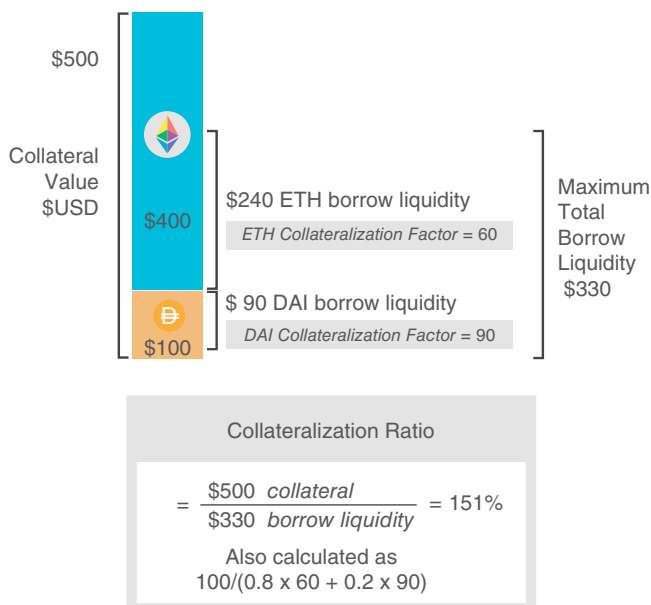


Figure 6.2 Collateralization Ratios in Compound

The formula for the borrow rate generally is an increasing linear function with a y intercept known as the *base rate* that represents the borrow rate at 0 percent borrow demand and a *slope* that represents the rate of change of the rates. These parameters are different for each ERC-20 asset supported by the platforms. Some markets have more advanced formulas that include a *kink*, which is a utilization ratio beyond which the slope steepens. These formulas can be used to reduce the cost of borrowing up to the kink and then increase the cost of borrowing after the kink to incentivize a minimum level of liquidity.

The supply interest rate is the borrow rate multiplied by the utilization ratio so that borrow payments can fully cover the

supplier rates. The *reserve factor* is a percentage of the borrow payments not given to the suppliers and instead set aside in a reserve pool that acts as insurance in case a borrower defaults. In an extreme price movement, many positions may become undercollateralized in that they have insufficient funds to repay the suppliers. In the event of such a scenario, the suppliers would be repaid using the assets in the reserve pool.

Here is a concrete example of the rate mechanics. In the DAI market, 100 million DAI is supplied, and 50 million is borrowed. Suppose the base rate is 1 percent and the slope is 10 percent. At 50 million borrowed, utilization is 50 percent. The borrow interest rate is then calculated to be $0.5 \cdot 0.1 + 0.01 = 0.06$, or 6 percent. The maximum supply rate (assuming a reserve factor of zero) would simply be $0.5 \cdot 0.06 = 0.03$, or 3 percent. If the reserve factor is set to 10, then 10 percent of the borrow interest is diverted to a DAI reserve pool, lowering the supply interest rate to 2.7 percent. Another way to think about the supply interest rate is that the 6 percent borrow interest of 50 million is equal to 3 million of borrow payments. Distributing 3 million of payments to 100 million of suppliers implies a 3 percent interest rate to all suppliers.

For a more complicated example involving a kink, suppose 100 million DAI is supplied, and 90 million DAI is borrowed – a 90 percent utilization. The kink is at 80 percent utilization, before which the slope is 10 percent and after which the slope is 40 percent, which implies the borrow rate will be much higher if the 80 percent utilization is exceeded.

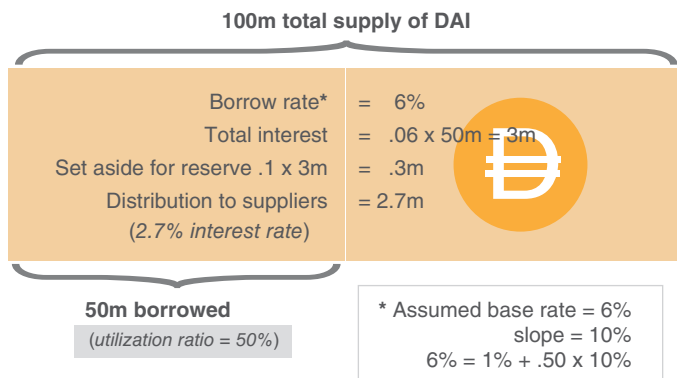


Figure 6.3 Savings and Lending Rates in Compound

The base rate remains at 1 percent. The borrow interest rate = 0.01 (base) + 0.8×0.1 (pre-kink) + 0.1×0.4 (post-kink) = 13 percent. The supply rate (assuming a reserve factor of zero) is $0.9 \times 0.13 = 11.7$ percent (Figure 6.3).

The utility of the Compound lending market is straightforward: it allows users to unlock the value of an asset without selling it and incurring a taxable event (at least under today's rules), similar to a home equity line of credit. Additionally, they can use the borrowed assets to engineer leveraged long or short positions, with competitive pooled rates and no approval process. For instance, if an investor is bearish on the price of ETH, they can simply deposit a stablecoin, such as DAI or USDC, as collateral and then borrow ETH and sell it for more of the stablecoin. If the price of ETH falls, investors use some of the DAI to purchase (cheaply) ETH to repay the debt. Compound offers several volatile

and stable tokens to suit the risk preferences of investors, and new tokens are continually added.

The Compound protocol must escrow tokens as a depositor to maintain that liquidity for the platform itself and to keep track of each person's ownership stake in each market. It would be naïve to keep track of the number inside a contract; instead, it would be better to tokenize the user's share. Compound does this using a cToken, one of the platform's important innovations.

Compound's cToken is an ERC-20 in its own right and represents an ownership stake in the underlying Compound market. For example, cDAI corresponds to the Compound DAI market, and cETH corresponds to the Compound ETH market. Both tokens are minted and burned in proportion to the funds added and removed from the underlying market as a means to track the amount belonging to a specific investor. Because of the interest payments that continually accrue to suppliers, these tokens are always worth more than the underlying asset. The benefit of designing the protocol in this way is that a cToken can be traded on its own like a normal ERC-20 asset. This trait allows other protocols to seamlessly integrate with Compound simply by holding cTokens and allows users to deploy their cTokens directly into other opportunities, such as using a cToken as collateral for a MakerDAO vault. Instead of using ETH only as collateral, an investors can use cETH and earn lending interest on the ETH collateral.

For example, assume there are 2,000 DAI in the Compound DAI market, and a total 500 cDAI represents the ownership in the market; this ratio of cDAI to DAI is not determinative

and could just as easily be 500,000 cDAI. At that moment in the example, 1 cDAI is worth 4 DAI, but after more interest accrues in the market the ratio will change. If a trader comes in and deposits 1,000 DAI, the supply increases by 50 percent (Figure 6.4). Therefore, the Compound protocol mints 50 percent more cDAI (250 cDAI) and transfers this amount to the trader's account. Assuming an interest rate of 10 percent, at year end there will be 3,300 DAI, and the trader's 250 cDAI can be redeemed for one-third, or 1,100,

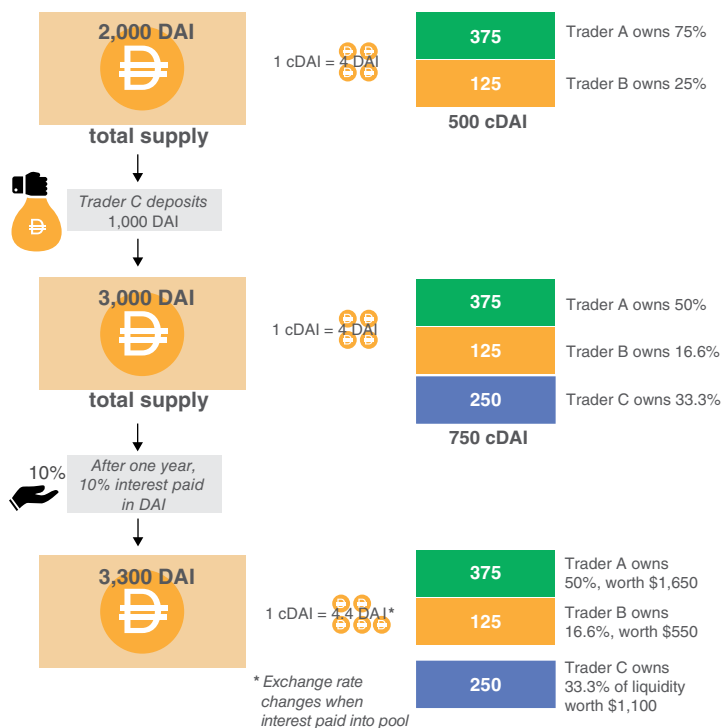


Figure 6.4 The Mechanics of Compound's Equity Token (cToken)

of the DAI. The trader can deploy cDAI in the place of DAI so that the DAI is not sitting idle but earning interest via the Compound pool. For example, the trader could deploy cDAI as the necessary collateral to open a perpetual futures position on dYdX or she could market make on Uniswap using a cDAI trading pair. (dYdX and Uniswap are discussed later in the chapter.)

The many different parameters of Compound's functionality – such as the *collateral factor*, *reserve factor*, *base rate*, *slope*, and *kink* – can all be tuned. The entity capable of tuning these parameters is Compound Governance, which has the power to change parameters, add new markets, freeze the ability to initiate new deposits or borrows in a market, and even upgrade some of the contract code itself. Importantly, Compound Governance cannot steal funds or prevent users from withdrawing. In the early stages of Compound's growth, governance was controlled by developer admins, similar to any tech startup. A strong development goal of Compound, as with most DeFi protocols, was to remove developer admin access and release the protocol to the leadership of a DAO via a governance token. The token allowed shareholders and community members to collectively become Compound Governance and propose upgrades or parameter tuning. A quorum agreement was required for any change to be implemented.⁷

Compound implemented this new governance system in May 2020 via the COMP token. COMP is used to vote on

protocol updates such as parameter tuning, addition of new asset support, and functionality upgrades (similar to MKR for MakerDAO). On June 15, 2020, the seventh governance proposal passed, which provided for distributing COMP tokens to users of the platform based on the borrow volume per market.⁸ The proposal offered an experience akin to a tech company giving its own stock to its users. The COMP token is distributed to both suppliers and borrowers and acts as a subsidization of rates. With the release of the token on public markets, COMP's market cap spiked to over \$2 billion. The price point of the distribution rate is so high that borrowing in most markets turned out to be profitable. This arbitrage opportunity attracted considerable volume to the platform, and the community governance has made and passed several proposals to help manage the usage.

The Compound protocol can no longer be turned off and will remain on Ethereum as long as Ethereum exists. Other platforms can easily escrow funds in Compound to provide additional value to their users or enable novel business models. An interesting example of this is Pool-Together,⁹ a no-loss lottery¹⁰ that deposits all user's funds into Compound but pays the entire pool's earned interest to a single random depositor at fixed intervals. Easy, instant access to yield or borrow liquidity on different Ethereum tokens makes Compound an important platform in DeFi (Table 6.2).

Table 6.2 Problems That Compound Solves

Traditional Finance Problem	Compound Solution
<i>Centralized control:</i> Borrowing and lending rates are controlled by institutions.	Compound rates are determined algorithmically and give control of market parameters to COMP stakeholders incentivized to provide value to users.
<i>Limited access:</i> Difficulty in accessing high-yield USD investment opportunities or competitive borrowing.	Open ability to borrow or lend any supported assets at competitive algorithmically determined rates (temporarily subsidized by COMP distribution).
<i>Inefficiency:</i> Suboptimal rates for borrowing and lending due to inflated costs.	Algorithmically pooled and optimized interest rates.
<i>Lack of interoperability:</i> Cannot repurpose supplied positions for other investment opportunities.	Tokenized positions via cTokens can be used to turn static assets into yield-generating assets.
<i>Opacity:</i> Unclear collateralization of lending institutions.	Transparent collateralization ratios of borrowers visible to entire ecosystem.

Aave

Aave¹¹ (launched in 2017) is a lending market protocol similar to Compound and offers several enhanced features. Aave offers many additional tokens to supply and borrow beyond what Compound offers. At the time of writing, Compound offers nine distinct tokens (different ERC-20 Ethereum-based assets), and Aave offers these nine plus an additional 13 not offered on Compound. Importantly, the Aave lending and variable borrowing rates are more predictable because, unlike the volatile COMP token in Compound, no subsidy is involved.

The Aave protocol supports the ability to create entirely new markets. Each market consists of its own group of token pools with their corresponding supply and borrow interest rates. The benefit of creating a separate market is that the market's supported tokens act as collateral solely in that market and cannot affect other markets, thus mitigating any potential contagion.

Aave currently has two main markets. The first is for more conventional ERC-20 tokens similar to those of Compound, supporting assets such as ETH, USDC, and DAI. The second is specific to Uniswap LP tokens. For example, when a user deposits collateral into a Uniswap market (known as a liquidity pool), they receive an LP token that represents their ownership in the market. The LP tokens can be deposited in the Uniswap market on Aave to generate additional returns.

Aave also supports flash loans in all of its markets and is the only source of flash liquidity for many smaller-cap tokens. Aave charges a fee of 9 basis points (bps) on the loan amount to execute a flash loan. The fee is paid to the asset pool and provides an additional return on investment to suppliers because they each own a pro rata share of the pool. An important use case for flash loans is that they allow users quick access to capital as a means to refinance positions. This functionality is crucial to DeFi, both as general infrastructure and as a component of a positive user experience (UX).

To provide an example, assume the price of ETH is 200 DAI. A user supplies 100 ETH in Compound and borrows 10,000 DAI to lever up and purchase an additional 50 ETH, which the user also supplies to Compound. Suppose the borrow interest rate in DAI on Compound is 15 percent, but only Aave is 5 percent. The goal is to refinance the borrowing to take advantage of the lower rate offered on Aave, which is analogous to refinancing a mortgage, a long and costly process in centralized finance.

One option is to manually unwind each trade on Compound and redo both trades on Aave to reconstruct the levered position, but this option is wasteful in terms of exchange fees and gas fees. The easier route is to take out a flash loan from Aave for 10,000 DAI, use it to pay the debt on Compound, withdraw the full 150 ETH, resupply to Aave, and trigger a normal Aave borrow position (at 5 percent APR) against that collateral to repay the flash loan (Figure 6.5). The latter approach effectively skips the

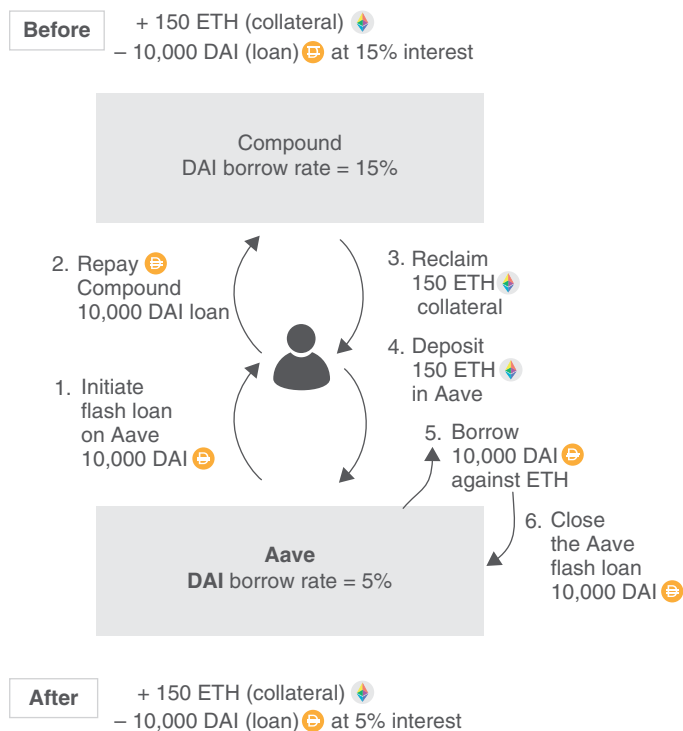


Figure 6.5 The Mechanics of an Aave Flash Loan

steps of exchanging ETH for DAI to unwind and rewind the leverage.

As shown in the preceding example, a flash loan used to refinance a position allows for DeFi client applications that let users migrate a levered position from one dApp to another with the single push of a button. These applications can even optimize portfolios for APR among several competing offerings, including Maker DSR (Dai Savings Rate), Compound, dYdX, and Aave.

An Aave innovation (and as of this writing available only on Aave) is a stable rate loan. (Using the label *stable* intentionally avoids calling it *fixed rate*.) A borrower has the option to switch between the variable rate and the current stable rate. The supply rate is always variable because under certain circumstances, such as if all borrowers left the market, it would be impossible to fund a fixed supply rate. The suppliers always collectively earn the sum of the stable and variable borrow interest payments minus any fees to the platform.

The stable rate is not a fixed rate because the rate is adjustable in extreme liquidity crunches and can be refinanced to a lower rate if market conditions allow. Also, some constraints exist around how much liquidity can be removed at a specific stable rate. Algorithmic stable borrowing rates provide value to risk-averse investors who wish to take on leverage without the uncertainty of a variable-rate position.

Aave is developing a *credit delegation* feature in which users can allocate collateral to potential borrowers who can use it to borrow a desired asset. Unsecured and reliant on trust, this process allows for uncollateralized loan relationships, such as in traditional finance, and potentially opens the floodgates in terms of sourcing liquidity. The credit delegation agreements will likely have fees and credit scores to compensate for the risk of unsecured loans. Ultimately, the delegator has sole discretion to determine who is an eligible

borrower and what contract terms are sufficient. Importantly, credit delegation terms can be mediated by a smart contract. Alternatively, the delegated liquidity can be given to a smart contract that can use the liquidity to accomplish its intended function. The underlying benefit of credit delegation is that all loans in Aave are ultimately backed by collateral, regardless of whose collateral it is.

For example, a supplier may have a balance of 40,000 DAI in Aave earning interest. The supplier wants to increase their expected return via an unsecured delegation of their collateral to a trusted counterparty. The supplier likely knows the counterparty through an off-chain relationship – perhaps a banking client. The counterparty can proceed to borrow, for instance, 100 ETH with the commitment to repay the asset to the supplier plus an agreed on interest payment. The practical impact is that the external relationship is unsecured because no collateral is available to enforce payment; the relationship is based essentially on trust.

In summary, Aave offers several innovations beyond the lending products offered by Compound and other competitors. Aave's flash loans, although not unique among competitors, provide additional yield to investors, making them a compelling mechanism to provide liquidity. These utilities also attract to the platform arbitrageurs and other applications that require flash liquidity for their use cases. Stable borrow rates are a key innovation, and Aave is the only platform currently with this offering. This feature could be

important for larger players who cannot operate under the potential volatility of variable borrow rates.

Finally, credit delegation allows users to unlock the value of supplied collateral in novel ways, including through traditional markets and contracts and even via additional layers of smart contracts that charge a premium rate to compensate for risk. Credit delegation allows loan providers to take their own collateral in the form of non-fungible Ethereum assets, perhaps tokenized art or real estate not supported by the main Aave protocol. As Aave continues to innovate, the platform will continue to amass more liquidity and cover a wider base of potential use cases (Table 6.3).

Table 6.3 Problems That Aave Solves

Traditional Finance Problem	Aave Solution
<i>Centralized control:</i> Borrowing and lending rates controlled by institutions.	Aave interest rates are controlled algorithmically.
<i>Limited access:</i> Only select groups have access to large quantities of money for arbitrage or refinancing.	Flash loans democratize access to liquidity for immediately profitable enterprises.
<i>Inefficiency:</i> Suboptimal rates for borrowing and lending due to inflated costs.	Algorithmically pooled and optimized interest rates.

(Continued)

Table 6.3 (Continued)

Traditional Finance Problem	Aave Solution
<i>Lack of interoperability:</i> Cannot monetize or utilize excess collateral in a lending position.	Credit delegation allows parties to use deposited collateral when they do not need borrowing liquidity.
<i>Opacity:</i> Unclear collateralization of lending institutions.	Transparent collateralization ratios of borrowers visible to the entire ecosystem.

DECENTRALIZED EXCHANGE

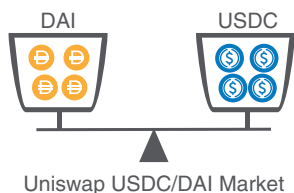
Uniswap

The primary example of an AMM on Ethereum is Uniswap.¹² We will focus our discussion on Uniswap v2. Recently, the third iteration of Uniswap was released and the v3 will be discussed later. Uniswap v2 uses a constant product rule to determine the trading price, using the formula $k = x * y$, where x is the balance of asset A , and y the balance of asset B . The product k is the *invariant* and is required to remain fixed at a given level of liquidity. To purchase (withdraw) some x , some y must be sold (deposited). The implied price is x/y and is the *risk-neutral* price because the contract is

equally willing to buy or sell at this rate as long as invariant k is constant.

Consider a concrete example (Figure 6.6). For simplicity, we will ignore transaction fees (gas) in all of the examples. Assume an investor in the Uniswap USDC/DAI market has 4 DAI (Asset A) and 4 USDC (Asset B). This sets the instantaneous exchange rate at 1 DAI:1 USDC and the invariant at 16 ($= x*y$). To sell 4 DAI for USDC, the investor deposits 4 DAI to the contract and withdraws 2 USDC. Now the USDC balance is $4 - 2 = 2$ and the DAI balance is $4 + 4 = 8$. The invariant remains constant at 16. Notice that the effective exchange rate is now 2 DAI: 1 USDC. The change in the exchange rate is due to slippage because of the low level of liquidity in the market. The magnitude of the invariant determines the amount of slippage. To extend the example, assume the balance is 100 DAI and 100 USDC in the contract. Now the invariant is 10,000, but the exchange rate is the same. If the investor sells 4 DAI for USDC, now 3.85 USDC can be withdrawn to keep the invariant constant and results in much lower slippage at an effective rate of 1.04 DAI: 1 USDC.

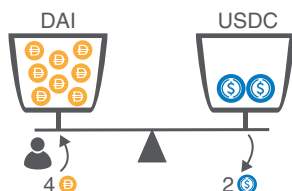
Deep liquidity helps minimize slippage. Therefore, it is important that Uniswap incentivizes depositors to supply capital to a given market. Anyone can become a liquidity provider by supplying assets on both sides of a market at the current exchange rate.¹³ Supplying both sides increases the product of the amount of assets held in the trading pair (i.e., increases the invariant as mentioned in the formula for



Instantaneous
exchange rate = 1 🟡 = 1 🔵
Invariant (K) = 4 🟡 $\times$ 4 🔵 = 16

Scenario A

Exchange 4 DAI

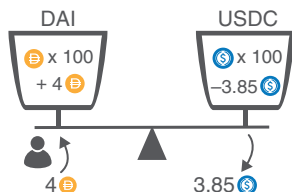


Invariant = $K = 8 \text{ 🟡} \times 2 \text{ 🔵} = 16$
Hence, 4 DAI exchanged for 2 USDC

Scenario B

Exchange 4 DAI

but contract has more liquidity, 100 DAI, 100 USDC



Instantaneous
exchange rate = 1 🟡 = 1 🔵
Before $K = 100 \times 100 = 10,000$
After $K = 104 \times 96.15 = 10,000$
Implied price = 1.04 🟡 = 1 🔵

Figure 6.6 The Mechanics of a Uniswap Automated Market Maker

the market maker). Following the preceding example, higher invariants lead to lower slippage and therefore an increase in effective liquidity. We can think of the invariant as a direct measure of liquidity. In summary, liquidity providing

increases the invariant with no effect on price, whereas trading against a market impacts the price with no effect on the invariant.

Each trade in a Uniswap market has an associated 0.3 percent fee that is paid back into the pool. Liquidity providers earn these fees based on their pro rata contribution to the liquidity pool and therefore prefer high-volume markets. This mechanism of earning fees is identical to the *cToken* model of Compound. The ownership stake is represented by a similar token called a UNI token. For example, the token representing ownership in the DAI/ETH pool is UNI DAI/ETH.

Liquidity providers in Uniswap essentially earn passive income in proportion to the volume on the market they are supplying. On withdrawal, however, the exchange rate of the underlying assets will almost certainly have changed. This shift creates an opportunity–cost dynamic (*impermanent loss*) that arises because the liquidity provider could simply hold the underlying assets and profit from the price movement. The fees earned from trading volume must exceed impermanent loss for liquidity providing to be profitable. Consequently, stablecoin trading pairs such as USDC/DAI are attractive for liquidity providers because the high correlation of the assets minimizes the impermanent loss.

Uniswap's $k = x * y$ pricing model works well if the correlation of the underlying assets is unknown. The model calculates the exact same slippage at a given liquidity level for any two trading pairs. In practice, however, we would expect much lower slippage for a stablecoin trading pair than for

an ETH trading pair because we know by design that the stablecoin's price should be close to \$1. The Uniswap pricing model leaves money on the table for arbitrageurs on high correlation pairs such as stablecoins because it does not adjust default slippage lower (change the shape of the bonding curve) as expected; the profit is subtracted from the liquidity providers. For this reason, competitor AMMs such as Curve¹⁴ that specialize in high-correlation trading pairs may cannibalize liquidity in these types of Uniswap markets.

If the pair does not already exist, anyone can start an ERC-20/ERC-20 or ETH/ERC-20 trading pair on Uniswap by simply supplying capital on both sides.¹⁵ The user determines the initial exchange rate, and arbitrageurs should drive that price to the true market price if it deviates at all. Users of the platform can effectively trade any two ERC-20 tokens supported by using *router contracts* that determine the most efficient path of swaps to get the lowest slippage if no direct trading pair is available.

A drawback of the AMM model is that it is particularly susceptible to front-running. This is not to be confused with illegal front-running, which plagues centralized finance. One of the features of blockchain is that all transactions are public. That is, when an Ethereum user posts a transaction to the memory pool, it is publicly visible to all Ethereum nodes. Front-runners can see this transaction – which is public information – and post a higher gas fee to trade against the pair before the user's transaction is added to a block; then they can immediately trade in the reverse direction against the pair. Estimates of front-running revenues, which come

directly at the expense of the users, grew from hundreds of thousands of dollars when front-running was first publicly demonstrated in 2017,¹⁶ to hundreds of millions of dollars as of mid-2021.¹⁷ Large transactions, especially in illiquid markets with high slippage, are particularly susceptible to front-running. For this reason, Uniswap allows users to set a maximum slippage as a clause in the transaction. If the acceptable level of slippage is exceeded, the trade will fail to execute.¹⁸ This provides a limit to the profit front-runners can make but does not completely remove the problem.

Another drawback is that arbitrage profits go only to arbitrageurs – they do not have a vested interest in the platform. The arbitrageurs profit at the expense of liquidity providers, who should not be losing the potential spread they would earn in a normal market-making scenario. Competing platforms, such as Mooniswap,¹⁹ propose to solve this issue by supplying virtual prices that slowly approach the true price, leaving tighter time windows and lower spreads for arbitrageurs to capitalize on. The additional spread remains in the pool for the liquidity providers.

Uniswap offers an interesting feature called a *flash swap*, which is similar to a flash loan. In a flash swap, the contract sends the tokens *before* the user pays for them with assets on the other side of the pair, unlocking many opportunities for arbitrageurs. The user can deploy this instant liquidity to acquire the other asset at a discount on another exchange before repaying it; the corresponding amount of the alternate asset must be repaid to maintain the invariant. This flexibility in a flash swap is different from the provision

in a flash loan, which requires that repayment occur with the same asset. A key aspect of a flash swap is that all trades must occur during a single Ethereum transaction and the trade must be closed with the corresponding amount of the complementary asset in that market.

Consider this example in the DAI/USDC market with a supply of 100,000 each (Figure 6.7). This implies a 1:1 exchange rate and an invariant of 10 billion. A trader who

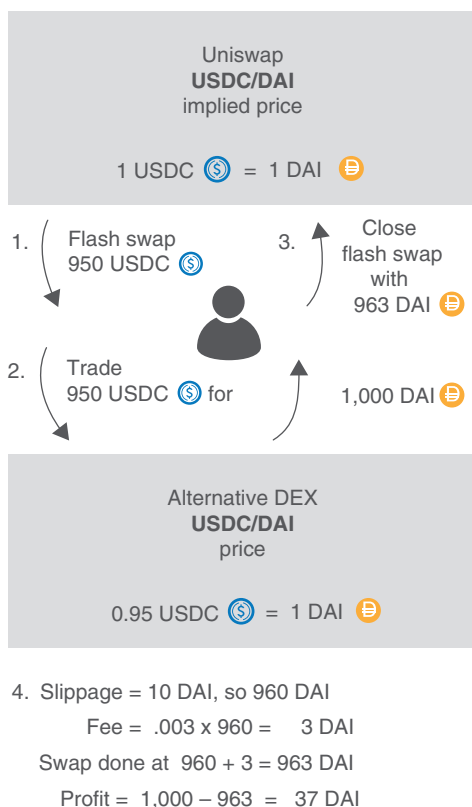


Figure 6.7 The Mechanics of a Flash Swap in Uniswap

has no starting capital spots an arbitrage opportunity to buy DAI on a DEX for 0.95 USDC. The trader can capitalize on this arbitrage via a flash swap by withdrawing 950 USDC of flash liquidity (liquidity derived from a flash loan) from the DAI/USDC market, purchase 1,000 DAI via the described arbitrage trade, and repay 963 DAI for a profit of 37 DAI – all consummated with no initial capital. The figure of 963 is calculated as 960 (with rounding for ease of illustration) to maintain the 10 billion invariant and to account for some slippage, plus a $0.30 \text{ percent} \times 960 = 3$ DAI transaction fee paid into the pool owned by the liquidity providers.

An important point about Uniswap is the release of a governance token in September 2020 called UNI. Like COMP, the Compound governance token, UNI is distributed to users to incentivize liquidity in key pools including ETH/USDC and ETH/DAI. The UNI governance even has some control over its own token distribution because 43 percent of the supply will be vested over four years to a treasury controlled by UNI governance. Importantly, each unique Ethereum address that had used Uniswap before a certain cutoff date (over 250,000 addresses) was given 400 UNI tokens as a free airdrop. At the same time as the airdrop, UNI was released on Uniswap and the Coinbase Pro exchange for trading. The price per token opened around \$3 with a total market cap of over \$500 million, amounting to \$1,200 of liquid value distributed directly to each user. This flood of supply could have led to selling pressure that tanked the token price. Instead, the token price

spiked to over \$8 before settling in the \$4–5 range. Through UNI, Uniswap effectively crowdsourced capital to build and scale its business, which attained a unicorn valuation for a short time. This demonstrates the value the community places in the token and the platform because the majority of supply is still held by those who received the airdrop.

As evidence that Uniswap is a good idea, it has been largely copied by Sushiswap.²⁰ Furthermore, the CFMM has been generalized by Balancer,²¹ in which more than two markets can be supported in a liquidity pool. In addition, the assets can be arbitrarily weighted (currently, Uniswap requires equal value).²² Further, the liquidity pool creator sets the transactions fees.

As of March 2021, the Uniswap team released a timeline and upgrade plan for the Uniswap protocol. Termed Uniswap v3, the Uniswap team proposed several changes to the protocol's liquidity provisioning model, moving away from the constant product formula described earlier and toward a model that resembles an on-chain, limit order book.²³ This change increases Uniswap's flexibility, allowing users and liquidity providers to customize curves and more actively manage their liquidity positions/control their return profiles. Uniswap v3 was launched May 5, 2021 and, as of writing, the volume of trading on v3 has already exceeded the v2 volume.²⁴

Uniswap is critical infrastructure for DeFi applications; it is important to have exchanges operational whenever it is needed. Uniswap offers a unique approach for generating

yield on users' assets by being a liquidity provider. The platform's flash swap functionality aids arbitrageurs in maintaining efficient markets and unlocks new use cases for users, who can access any ERC-20 token listed, including creating completely new tokens through an IDO. As AMM volume grows on Ethereum and new platforms arise with competing models, Uniswap will continue to be a leader and an example of critical infrastructure going forward (Table 6.4).

Table 6.4 Problems That Uniswap Solves

Traditional Finance	
Problem	Uniswap Solution
<i>Centralized control:</i> Exchanges that control which trading pairs are supported.	Allows anyone to create a new trading pair if it does not already exist and automatically routes trades through the most efficient path if no direct pair exists.
<i>Limited access:</i> The best investment opportunities and returns from liquidity providing are restricted to large institutions.	Anyone can become a liquidity provider and earn fees for doing so. Any project can list its token on Uniswap to give anyone access to an investor.
<i>Inefficiency:</i> Trades generally require two parties to clear.	An AMM that allows constant access for trading against the contract.

(Continued)

Table 6.4 (Continued)

Traditional Finance	
Problem	Uniswap Solution
<i>Lack of interoperability:</i> Ability to exchange assets on one exchange is not easily used within another financial application.	Any token swap needed for a DeFi application can utilize Uniswap as an embedded feature.
<i>Opacity:</i> Unknown if the exchange truly owns all user's entire balance.	Transparent liquidity levels in the platform and algorithmic pricing.

DERIVATIVES

Yield Protocol

Yield Protocol²⁵ proposes a derivative model for secured, zero-coupon bonds. Essentially, the protocol defines a *yToken* to be an ERC-20 (fungible) token that settles in some fixed quantity of a target asset at a specified date. The contract will specify that the tokens – which have the same expiry, target asset, collateral asset, and collateralization ratio – are fungible. They are secured by the collateral asset and have a required maintenance collateralization ratio similar to, for example, MakerDAO, and to other DeFi platforms we have already discussed. If the collateral's value dips below

the maintenance requirement, the position can be liquidated with some or all of the collateral sold to cover the debt.

The mechanism for yToken settlement is still undecided, but one proposed solution is “cash” settlement, which means paying an equivalent amount of the collateral asset worth the specified amount of the target asset. For example, if the target asset is 1 ETH secured by 300 DAI, and at expiry $1 \text{ ETH} = 200 \text{ DAI}$, a cash settlement would pay out 200 DAI and return the 100 DAI excess collateral to the seller of the yToken. The other commonly proposed solution is “physical” settlement, which automatically sells collateral for the target asset upon expiry (perhaps on Uniswap) to pay out in the target asset. Using the same numbers as in the previous example, the owner of the yToken would receive 1 ETH and the seller would receive slightly less of the remaining collateral, likely around 95 DAI, after subtracting exchange fees. The yToken effectively allows for fixed-rate borrowing and lending, using the implied return on the discounted price of the token versus the target amount.

We can illustrate as follows: assume a user has a yToken with the target asset of 1 DAI backed by ETH. The maturity date is one year ahead, and the yToken is trading at 0.92 DAI. A purchase of the yToken effectively secures an 8.7 percent fixed interest rate, even in the case of a liquidation. In the event of a normal liquidation, the collateral would be sold to cover the position, as shown in Figure 6.8.

A compelling third option for settlement (in addition to cash and physical) is “synthetic” settlement. Here, the

DeFi Deep Dive

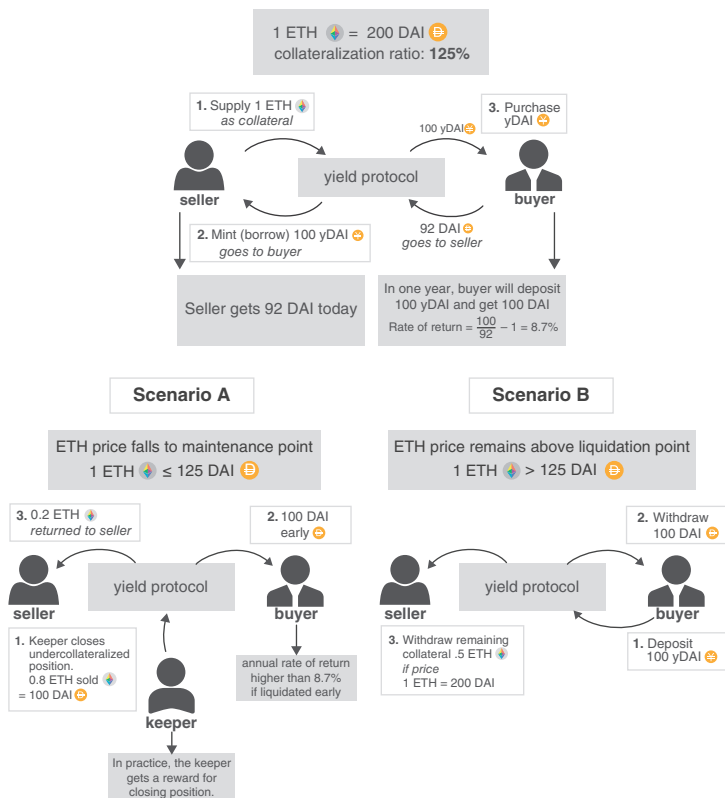


Figure 6.8 The Mechanics of Fixed-Rate Borrowing in the Yield Protocol

underlying asset is not directly repaid but instead is rolled into an equivalent amount of that asset pool on a lending platform such as Compound. Synthetic settlement means that yDAI could settle in cDAI, converting the fixed rate into a floating rate. Buyer could close the position and redeem cDAI for DAI at their leisure. The Yield Protocol

handles all these conversions for users so that user experience simply revolves around the target asset.

In the Yield Protocol white paper,²⁶ the authors discuss interesting applications from the investor's perspective. An investor can purchase yTokens to synthetically lend the target asset. The investor would be paying X amount of the asset now to purchase the yTokens. Upon settlement, the investor receives $X + \text{interest}$. This financial transaction in total is functionally a lend of the target asset. Note that the interest is implied in the pricing and not a directly specified value. Alternatively, yTokens can be minted and sold to synthetically borrow the target asset, meaning X amount of the asset is received now (the face value) with the promise to pay $X + \text{interest}$ in the future. This financial transaction is functionally a borrow of the target asset.

Additional applications include a perpetual product on top of yTokens that maintains a portfolio of different maturities and rolls short-term profits into long-term yToken contracts. For example, the portfolio may include three-, six-, and nine-month plus one-year maturity yTokens; once the three-month tokens mature, the smart contract can reinvest the balance into one-year maturity yTokens. Token holders in this fund would essentially be experiencing a floating rate yield on the underlying asset with rate updates every three months. The yTokens also allow for the construction of yield curves by analyzing the implied yields of short and longer term contracts. This allows observers to quantify investor sentiment among the various supported target assets.

The Yield Protocol can even be directly used to speculate on interest rates. Several DAI derivative assets – Compound cDAI, Aave aDAI, and Chai²⁵ – represent a variable interest rate. One can imagine a seller of yDAI using one of these DAI derivative assets as collateral. The effect of this transaction is that the seller is paying the fixed rate on the yDAI while receiving the variable rate on the collateral. This is a bet that rates will increase. Likewise, purchasing yDAI (of any collateral type) is a bet that variable rates will not increase beyond the fixed rate received.

Yield is an important protocol that supplies fixed rate products to Ethereum. It can be tightly integrated with other protocols like MakerDAO and Compound to create robust interest-bearing applications for investors. Demand for fixed income components will grow as mainstream investors begin adopting DeFi with portfolios in need of these types of assets (Table 6.5).

Table 6.5 Problems That the Yield Protocol Solves

Traditional

Finance Problem

Yield Solution

Centralized control:
Fixed income
instruments largely
restricted to governments
and large corporations.

Yield protocol is open to parties
of any size.

(Continued)

Table 6.5 (Continued)

Traditional	
Finance Problem	Yield Solution
<i>Limited access:</i> Many investors have limited access to buy or sell sophisticated fixed income investments.	Yield allows all market participants to buy or sell a fixed income asset that settles in a target asset of their choosing.
<i>Inefficiency:</i> Fixed income rates are lower due to layers of fat in traditional finance.	Lean infrastructure running on Ethereum allows for more competitive rates and diverse liquidity pools due to the elimination of middlemen.
<i>Lack of interoperability:</i> Fixed income instruments generally settle in cash that the investor must determine how to allocate.	yTokens can settle in any Ethereum target asset and even settle synthetically into a floating-rate lending protocol to preserve returns.
<i>Opacity:</i> Risk and uncertainty of counterparty in traditional agreements.	Clear collateralization publicly known on Ethereum blockchain backing the investment.

dYdX

dYdX²⁶ specializes in derivatives and margin trading, which currently supports a variety of cryptocurrencies in addition to ETH and BTC. The company has a spot DEX that allows

investors to exchange these assets against the current bid–ask on the order book and uses a hybrid on–off chain approach. Essentially, dYdX stores *signed*, or preapproved, orders without submitting to Ethereum; they use cryptography to guarantee they are used only to exchange funds for the desired asset at the desired price. The DEX supports limit orders and a *maximum slippage* parameter for market orders in an effort to mitigate the slippage associated with price moves or front-running.

dYdX provides market makers and traders the open-source software and a user interface required to interact with the DEX. Having dYdX do the order matching introduces a certain element of trust because the infrastructure could be in downtime or not posting transactions for some reason. Allowing dYdX to match the orders holds little or no risk that the company could steal user funds because the signed orders can be used only as intended per the smart contract. When the orders are matched, they are submitted to the Ethereum blockchain, where the smart contract facilitates settlement.

In addition, an investor can take a levered long or short position up to 10 times using margined collateral. The positions can be isolated so that a single collateral deposit is used or cross-margined to pool investor's balances for collateral. As in other protocols, dYdX has a maintenance margin requirement that, if not maintained, triggers liquidation of the collateral to close the position. The liquidations can be performed by external keepers who are paid to find and

liquidate underwater positions, similar to the process followed by MakerDAO.

dYdX offers borrowing and lending similar to Compound and Aave. It also features free flash loans (Aave's are not free), which makes it a popular choice for DAI, ETH, and USDC flash liquidity. In the world of open smart contracts, it makes sense that flash loans rates would be driven to zero since they are nearly risk-free. Lending rates are determined by the loan's duration and relative risk of default. For flash loans, repayment is algorithmically enforced, and time is infinitesimal. In a single transaction, only the user can make any function calls or transfers; no other Ethereum users can move funds or make any changes while a particular user's transaction is in flight, resulting in no opportunity cost for the capital. Hence, as expected, a market participant offering free flash loans will attract more usage to their platform. Because flash loans do not require any up-front capital, they democratize access to funds for various use cases. In the Aave example, we show how flash loans can be used to refinance a loan. We will now illustrate the use of flash loans to capitalize on an arbitrage opportunity.

Suppose the effective exchange rate for 1,000 DAI for ETH on Uniswap is 6 ETH/1,000 DAI. (The instantaneous exchange rate would be different due to slippage.) Also, suppose the dYdX DEX has a spot ask price of 5 ETH for 1,000 DAI (i.e., the ETH are much more expensive on

dYdX than Uniswap). To capitalize on this arbitrage opportunity, without any capital beyond the gas fee, an investor can execute a flash loan to borrow 1,000 DAI, exchange it on Uniswap for 6 ETH, and use 5 of those ETH to trade for 1,000 DAI on dYdX. Finally, the investor can repay the flash loan with the 1,000 DAI and pocket the 1 ETH profit. This all happens in a single transaction; multiple contract executions can happen in a single transaction on the Ethereum blockchain (Figure 6.9).

The main derivative products dYdX offers are ETH and BTC perpetual futures. At the time of writing, dYdX also

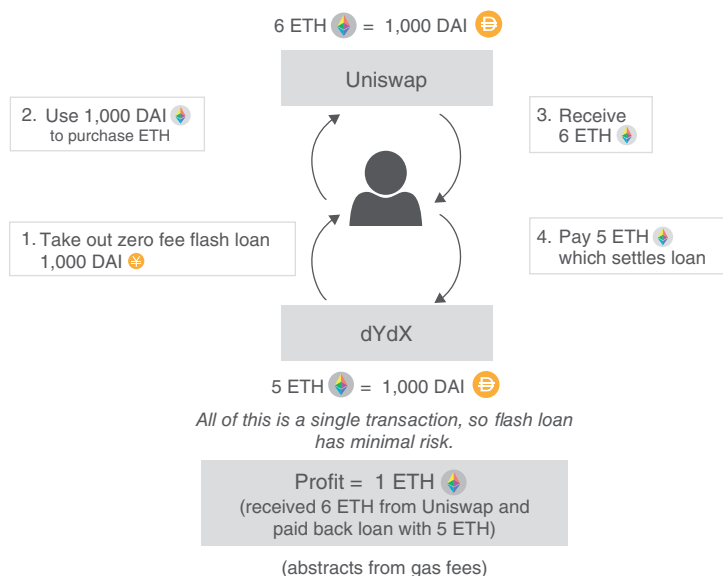


Figure 6.9 The Mechanics of Arbitrage with dYdX

offered 11 other cryptocurrency futures. A perpetual futures contract is similar to a traditional futures contract but without an expiration date. By entering into a perpetual futures contract, the investor is simply betting on the future price of an asset. The contract can be long or short and with or without leverage; it uses an index price based on the average price of the underlying asset across the major exchanges.²⁷ The investor deposits margin collateral and chooses a direction and amount of leverage. Depending on investor demand, the contract can trade at a premium or discount to the index price (BTC).

A funding rate, paid from one side to the other, keeps the futures price close to the index. If the futures contract is trading at a premium to the index, the funding rate would be positive and longs would pay shorts. The magnitude of the funding rate is a function of the difference in price compared with the index. Likewise, if the contract is trading at a discount, the shorts pay the long positions. The funding rate incentivizes investors to take up the opposing side from the majority to keep the contract price close to the index.²⁸ As long as the required margin is maintained, the investor can always close the position at the difference in the price of the notional position minus any negative balance held on margin.

Like a traditional futures contract, the perpetual futures contract has two margins: initial and maintenance. Suppose the initial margin is 10 percent. This means the investor

needs to post collateral (or equity) worth 10 percent of the underlying asset. A long futures contract allows the investor to buy the asset at a set price in the future. If the market price rises, the investor can buy the asset at a price cheaper than the market price and the profit is the difference between the market price and the contract price. A short position works similarly except that the investor agrees to sell the asset at a fixed price. If the market price falls, the investor can purchase the asset in the open market and sell at the higher price stipulated in the contract. The profit is the difference between the contract price and the market price.

The risk is that the price moves against the investor. For example, if the investor is long with a 10 percent margin and the market price drops by 10 percent, the collateral is gone because the difference between purchasing at the contract price and selling in the open market (at a loss) wipes out the value of the collateral. Importantly, futures are different from options. If the underlying asset's price moves the wrong way in an option contract, the option holder can walk away: the exercise of the option is discretionary – that is why it is called an option – and no trader would exercise an option to guarantee a loss. Futures, however, are obligations. As such, traditional exchanges have mechanisms that seek to minimize the chance the contract holder does not default on a losing position.

The maintenance margin is the main tool to minimize default. Suppose the maintenance margin is 5 percent.

On a traditional futures exchange, if the price drops by 5 percent investors are required to replenish the collateral to bring it back up to 10 percent. If investors fail to do this, the exchange liquidates the position. A similar mechanism exists on dYdX, but with important differences. First, if any position falls to 5 percent, keepers will trigger liquidation. If any collateral remains, they may keep it as a reward. Second, the liquidation is almost instantaneous. Third, no centralized exchange exists. Fourth, dYdX contracts are perpetual, whereas traditional exchange contracts usually have a fixed maturity date.²⁹

Consider the following example. Suppose the BTC price index is 10,000 USDC/BTC. An investor initiates a long position by depositing 1,000 USDC as margin (collateral), creating a levered bet on the price of BTC. If the price rises by 5 percent, the profit is 500. Given the investor has only deposited 1,000, the investor's rate of return is 50 percent, or $(1,000 - 500)/1,000$.

We can also think about the mechanics another way. Taking a long position at 10,000, the investor is committing to buying at 10,000 and the obligation is 10,000. Think of the obligation as a negative balance because the investor must pay 10,000 according to the contract. The investor has already committed collateral of 1,000 and owes 9,000. On the other side, the investor has committed those funds to purchase an asset, 1 BTC. The investor thus has a positive balance

of 10,000, the current price. The collateralization ratio is $10,000/9,000 = 111$ percent, which is a margin percentage of 11 percent and is nearly the maximum amount of allowed leverage (10 percent margin).

This intuition works similarly for a short position. The investor has committed to sell at 10,000, which is a positive balance and is supplemented by the margin deposit of 1,000 (so total of 11,000). The investor's negative balance is the obligation to buy 1 BTC, currently worth 10,000. The collateralization ratio is $11,000/10,000$, which corresponds to a margin of 10 percent.

Let's now follow the mechanics of a short position when the underlying asset (BTC) increases in value by 5 percent. If the price of BTC increases to 10,500 (a 5 percent increase), the margin percentage becomes $(11,000/10,500) - 1 = 4.76$ percent and the short position becomes subject to liquidation. The net balance of the position is \$500, the incentive for the liquidator to close the position collect the balance. Figure 6.10 reviews the mechanics of a long position.

The dYdX BTC perpetual futures contract allows investors to access BTC returns natively on the Ethereum blockchain while being able to supply any ERC-20 asset as collateral. Perpetual futures are rising in popularity, and this functionality may continue to attract liquidity over time.

DeFi and the Future of Finance

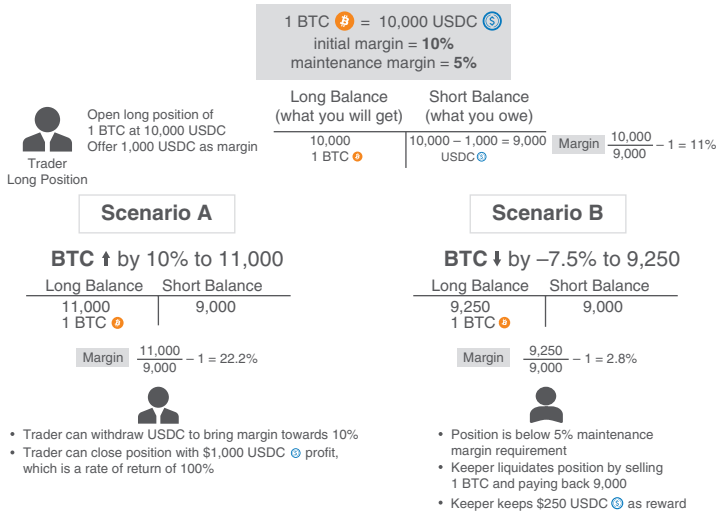


Figure 6.10 Perpetual Futures with dYdX

Table 6.6 Problems That dYdX Solves

Traditional Finance

Problem

Centralized control: Borrowing and lending rates controlled by institutions.

dYdX Solution

dYdX rates are determined algorithmically based on clearly outlined, transparent formulas (often asset pool utilization rates).

(Continued)

Table 6.6 (Continued)

Traditional Finance	
Problem	dYdX Solution
<i>Limited access:</i> Difficulty in accessing high yield USD investment opportunities or competitive borrowing as well as futures and derivative products. Access to capital for immediately profitable enterprises is limited.	Open ability to borrow or lend any supported assets at competitive algorithmically determined rates. Includes a perpetual futures contract that could synthetically support any asset. Free flash loans give developers access to large amounts of capital to capitalize on arbitrage or other profitable opportunities.
<i>Inefficiency:</i> Suboptimal rates for borrowing and lending due to inflated costs.	Algorithmically pooled and optimized interest rates. Free flash loans offered for immediate use cases.
<i>Lack of interoperability:</i> Difficult to repurpose funds within a financial instrument.	Flash loans can immediately utilize the entirety of the AUM for outside opportunities without risk or loss to investors.
<i>Opacity:</i> Unclear collateralization of lending institutions.	Transparent collateralization ratios of borrowers are visible to the entire ecosystem.

Synthetix

Many traditional derivative products have a decentralized counterpart. DeFi, however, allows new types of derivatives because of smart contracts. Synthetix³⁰ is developing such a new type of derivative.

Imagine creating a derivative cryptoasset, whose value is based on an underlying asset that is neither owned nor escrowed. Synthetix is one group whose primary focus is creating a wide variety of liquid synthetic derivatives. Its model is, at a high level, straightforward and novel. The company issues *Synths*, tokens whose prices are pegged to an underlying price feed and are backed by collateral. MakerDAO's DAI is also a synthetic asset. The price feeds come from the Chainlink's³¹ decentralized oracles.³² Synths can theoretically track any asset, long or short, and even levered positions. In practice, there is no leverage, and the main tracked assets are cryptocurrencies, fiat currencies, and gold.

A long Synth is called an *sToken*, for example, a sUSD or a sBTC. The sUSD is a synthetic because its value is based on a price feed. A short Synth is called an *iToken*, for example, an iETH or an iMKR. Synthetix also has a platform token called SNX. SNX is not a governance token like MKR and COMP, but is a *utility token* or a *network token*, which means it enables the use of Synthetix functionality as its only feature. SNX serves as the unique collateral asset for the entire system. When users mint Synths against their SNX, they incur a debt proportioned to the total outstanding debt denominated in USD. They become *responsible* for this percentage

of the debt in the sense that to unlock their SNX collateral they need to return the total USD value of their debt. The global debt of all Synths is thus shared collectively by the Synth holders based on the USD-denominated percentage of the debt they owned when they opened their positions. The total outstanding USD-denominated debt changes when any Synth's price fluctuates, and each holder remains responsible for the same percentage they were responsible for when they minted their Synths. Therefore, when a SNX holder's Synths outperform the collective pool, the holder effectively profits, and vice versa, because their asset value (their Synth position) outpaced the growth of the debt (sum of all sUSD debt).

As an example, three traders each have \$20,000 for a total debt of \$60,000: one holds 2 sBTC priced at \$10,000 each, one holds 100 sETH priced at \$200 each, and one holds 20,000 sUSD priced at \$1 each. Each has a debt proportion of 33.3 percent. If the price of BTC doubles to \$20,000 and the price of ETH spikes to \$1,000, the total debt becomes $\$160,000 = \$40,000 \text{ (sBTC)} + \$100,000 \text{ (sETH)} + \$20,000 \text{ (sUSD)}$.³³ Because each trader is responsible for 33.3 percent, about \$53,300, only the sETH holder is profitable even though the price of BTC doubled. If the price of BTC falls to \$5,000 and ETH to \$100, then the total debt falls to \$40,000 and the sUSD holder becomes the only profiting trader. Figure 6.11 details this example.

The platform has a native DEX that will exchange any two Synths at the rate quoted by the oracle. Traders pay the exchange fees to a fee pool redeemable by SNX holders in

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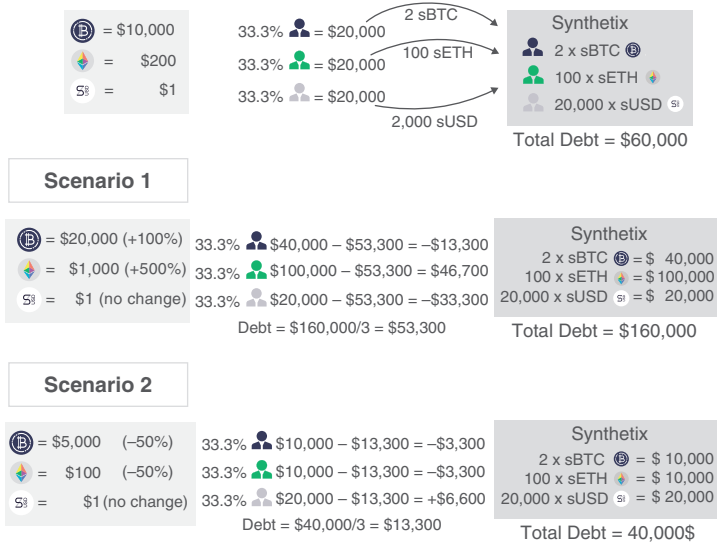


Figure 6.11 The Mechanics of Synthetix

proportion to their percentage of the debt. The contracts enforce that SNX holders can redeem their fees only if they maintain a sufficient collateralization ratio relative to their portion of the debt. The required collateralization ratio to mint Synths and participate in staking rewards is high, currently 750 percent. The Synthetix protocol also mints new SNX tokens via inflation to reward various stakeholders in the ecosystem for contributing value. The protocol distributes the rewards as a bonus incentive for maintaining a high collateralization ratio or increasing the liquidity of SNX.

As of writing there were 36 crypto synths available for trading and seven foreign currency synths. Synthetix also allows users to trade certain equities as well as gold and oil. The protocol is also beginning to offer a binary options

trading interface, further expanding its capabilities. The platform could easily gain popularity because there is no slip-page against the price feed; however, the pooled liquidity and shared debt models offer interesting challenges (Table 6.7).

Table 6.7 Problems That Synthetix Solves

Traditional

Finance Problem	Synthetix Solution
<i>Centralized control:</i> Assets can generally only be bought and sold on registered exchanges.	Offer synthetic assets in one place that can track any real world asset.
<i>Limited access:</i> Access to certain assets is geographically limited.	Anyone can access Synthetix to buy and sell Synths. Some restrictions may eventually apply to Synths that are securities.
<i>Inefficiency:</i> Large asset purchases suffer from slippage as traders eat into the liquidity pool.	Synths exchange rates are backed by a price feed, which eliminates slippage.
<i>Lack of interoperability:</i> Real-world assets such as stocks can't be easily represented directly on a blockchain	Synth representations of real assets are totally compatible with Ethereum and other DeFi protocols.
<i>Opacity:</i> Lack of transparency in traditional derivative markets.	All protocol-based projects and features are transparently funded and voted upon by a DAO

TOKENIZATION

Tokenization refers to the process of taking some asset or bundle of assets, either on or off chain, and

1. representing that asset on chain with possible fractional ownership; or
2. creating a composite token that holds some number of underlying tokens.

A token can conform to different specifications based on the type of properties a user wants the token to have. As mentioned earlier, the most popular token standard is ERC-20, the fungible token standard. This interface defines abstractly how a token, which has units that are non-unique and interchangeable (such as USD), should behave. An alternative is the ERC-721 standard, which defines non-fungible tokens (NFTs). These tokens are unique, such as a token representing ownership of a piece of fine art or a specific digital asset from a game. DeFi applications can take advantage of these and other standards to support any token using the standard simply by coding for the single standard.

Set Protocol

Set Protocol³⁴ offers the “composite token” approach to tokenization. Instead of tokenizing assets non-native to Ethereum, Set Protocol combines Ethereum tokens into composite tokens that function more like traditional

exchange traded funds (ETFs). Set Protocol combines cryptoassets into *Sets*, which are ERC-20 tokens and fully collateralized by the components escrowed in a smart contract. A Set token is always redeemable for its components. Sets can be static or dynamic, based on a trading strategy. Static Sets are straightforward to understand and are simply bundled tokens the investor cares about; the resulting Set can be transferred as a single unit.

Dynamic Sets define a trading strategy that determines when reallocations can be made and at what times. Some examples include the “Moving Average” Sets that shift between 100 percent ETH and 100 percent USDC whenever ETH crosses its X-day simple or exponentially weighted moving average. Similar to normal ETFs, these Set tokens have fees and sometimes performance-related incentives. At the Set’s creation, the manager pre-programs the fees, which are paid directly to the manager for that particular Set. The available fee options are a buy fee (front-end load fee), a streaming fee (management fee), and a performance fee (percentage of profits over a high-water mark). The Set Protocol currently takes no fee for itself, although it may add a fee in the future. The prices and returns for Set Protocol are calculated via MakerDAOs’ publicly available oracle price feeds, which are also used by Synthetix. The main value-add of dynamic Sets is that the trading strategies are publicly encoded in a smart contract so users know exactly how their funds are being allocated and can easily redeem at any time.

Set Protocol also has a *Social Trading* feature in which a user can purchase a Set whose portfolio is restricted to certain assets with reallocations controlled by a single trader. Because these portfolios are actively managed, they function much more like active mutual funds or hedge funds. The benefits are similar in that the portfolio manager has a predefined set of assets to choose from, and the users benefit from this contract-enforced transparency.

For example, a portfolio manager for a Set has a goal to “buy low and sell high” on ETH. The only assets they can use are ETH and USDC, and the only allocations they are allowed are 100 percent ETH and 100 percent USDC. At their sole discretion, they can trigger a contract function to rebalance the portfolio entirely into one asset or the other; this is the only allocation decision they can make. Assume they start with 1,000 USDC. The price of ETH dips to 100 USDC/ETH and they decide to buy. They can trigger a rebalance to have 10 ETH in the Set. If the price of ETH doubles to \$200, the entire Set is now worth \$2,000. A shareholder who owns 10 percent of the Set can redeem their shares for 1 ETH.

Sets could democratize wealth management in the future by being more peer to peer, allowing fund managers to gain investment exposures through non-traditional channels, and giving all investors access to the best managers. A further enhancement many Sets take advantage of is that their components use cTokens, the Compound-invested version of tokens. Between rebalances, tokens earn interest through

the Compound protocol. This is one example of DeFi platforms being composed to create new products and value for investors.

Table 6.8 Problems That Set Protocol Solves

Traditional	
Finance Problem	Set Protocol Solution
<i>Centralized control:</i> Fund managers can control their funds against the will of investors.	Enforces sovereignty of the investor over their funds at the smart contract level.
<i>Limited access:</i> Talented fund managers often are unable to gain exposures and capital to run a successful fund.	Allows all to become fund managers and display their skills using social trading features.
<i>Inefficiency:</i> Many arising from antiquated practices.	Trading strategies encoded in smart contracts lead to optimal execution.
<i>Lack of interoperability:</i> Difficult to combine assets into new packages and incorporate the combined assets into new financial products.	Set tokens are ERC-20 compliant tokens that can be used on their own in other DeFi protocols. For example, Aave allows Set token borrowing and lending for some popular Sets.
<i>Opacity:</i> Difficult to know the breakdown of assets in an ETF or mutual fund at any given time.	Total transparency into strategies and allocations of Set tokens.

Wrapped Bitcoin

The wrapped bitcoin (wBTC)³⁵ application takes the *representing off-chain assets on chain* approach to tokenization, specifically for BTC. Abstractly, wBTC allows BTC to be included as collateral or liquidity on all of the Ethereum-native DeFi platforms. Given that BTC has comparatively low volatility³⁶ and is the most well-adopted cryptocurrency by market cap, this characteristic unlocks a large potential capital pool for DeFi dApps.

The wBTC ecosystem contains three key stakeholders: users, merchants, and custodians. Users are simply the traders and DeFi participants who generate demand for the value proposition associated with wBTC, namely, Ethereum-tokenized BTC. Users can purchase wBTC from merchants by transferring BTC and performing the requisite KYC/AML, thus making the entry and exit points of wBTC centralized and reliant on off-chain trust and infrastructure. Merchants are responsible for transferring BTC to the custodians. At the point of transfer, the merchant signals to an on-chain Ethereum smart contract that the custodian has taken custody of the BTC and is approved to mint wBTC. Custodians use industry-standard security mechanisms to custody the BTC until it is withdrawn from the wBTC ecosystem. Once the custodians have confirmed receipt, they can trigger the minting of wBTC that releases wBTC to the merchant. Finally, closing the loop, the merchant transfers the wBTC to the user.

No single participant can control the minting and burning of wBTC, and all BTC entering the system is audited via transaction receipts that verify custody of on-chain funds. These safeguards increase the system's transparency and reduce the risk to users that is inherent in the system. Because the network consists of merchants and custodians, any fraud is quickly expungable from the network at only a small overall cost versus the cost that would be incurred in a single centralized entity. The mechanism by which merchants and custodians enter and leave the network is a multi-signature wallet controlled by the wBTC DAO. In this case, the DAO does not have a governance token; instead, a set of owners who can add and remove owners controls the DAO. The contract currently allows a maximum of 50 owners, with a minimum threshold of 11 to invoke a change. The numbers 50 and 11 can be changed, if a number of conditions are met. This system is more centralized than other governance mechanisms we have discussed, but is still more decentralized than allowing a single custodian to control all the wBTC.